2.1 T/F Questions

1-5: FTTTF, 6-10: FTFFT, 11-15: TFFFT, 16-20: FTFFT

2.3. Standard Questions

2.3.1

Find a product of cyclic groups that is isomorphic to the group

$$(\mathbb{Z}_{12} \times \mathbb{Z}_{12})/\langle (2,6) \rangle.$$

Solution: $\mathbb{Z}_{12} \times \mathbb{Z}_{12}$ is generated by $(1,3)$ and $(1,2)$, so $\mathbb{Z}_{12} \times \mathbb{Z}_{12}/\langle (2,6) \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_{12}$.

2.3.2

Let $\phi : G \to H$ be a homomorphism of groups. Let K be a subgroup of $\mathrm{Im}(\phi)$. Show that

$$N_G\Big(\phi^{-1}(K)\Big) = \phi^{-1}\Big(N_H(K)\Big).$$

Proof: If $g \in N_G(\phi^{-1}(K))$ then for any $k \in K \subset \mathrm{Im}\phi$, take $h \in G$ such that $\phi(h) = k$, then $ghg^{-1} \in \phi^{-1}(K) \implies \phi(g)k\phi(g)^{-1} \in K$ so $g \in \phi^{-1}(N_H(K))$.
If $g \in \phi^{-1}(N_H(K))$ then for any $h \in G$ such that $\phi(h) = k \in K$, $\phi(g)k\phi(g)^{-1} \in K$ implies $ghg^{-1} \in \phi^{-1}(K)$ so $g \in N_G(\phi^{-1}(K))$.

2.3.3

Let G and H be two groups. Suppose that there is an injective homomorphism $i : H \to G$ and a homomorphism $\pi : G \to H$ such that

$$\pi \circ i = \mathrm{id}_H \,.$$

Show that one can write G as a semidirect product $H \rtimes \ker(\pi)$ such that i is the embedding of H into the first factor, and π is the projection to the first factor (by quotienting out the normal subgroup $\ker(\pi)$).
Can you give an example where this semidirect product is not a direct product?
Proof: Let $N = \mathrm{Ker}\pi$ then $H \cong G/N$. Consider $\phi : G \to H \rtimes N, g \mapsto (\pi(g), g\iota(\pi(g))^{-1})$ then $\phi(g) = 1$ implies $\pi(g) = 1$ and $g = \iota(\pi(g)) = 1$ so ϕ is injective. For each $h \in H, n \in N$, let $g = n\iota(h)$ then $\phi(g) = (h, n)$ so ϕ is surjective. Hence ϕ gives an isomorphism $G \cong H \rtimes N$.
Consider $\mathrm{sgn} : S_3 \to \mathbb{Z}_2$ and $\iota : \mathbb{Z}_2 \to S_3, 1 \mapsto (12)$ then $S_3 \neq \mathbb{Z}_2 \times A_3$ which is abelian.

2.3.4

Let

$$G = \langle x_1 \rangle \times \langle x_2 \rangle \times \cdots \times \langle x_n \rangle$$

be a finite abelian group (written in multiplicative convention) with $|x_i| = n_i$. Consider the pth-power map

$$\phi : G \longrightarrow G, \qquad x \longmapsto x^p.$$

1. Prove that ϕ is a homomorphism.
2. Describe the image and the kernel of ϕ in terms of the given generators. (The answer depends on whether each n_i is divisible by p.)
3. Prove that both $\ker(\phi)$ and $G/\operatorname{im}(\phi)$ are elementary p-groups, namely products of copies of $\mathbb{Z}/p\mathbb{Z}$, and that they contain the same number of copies of $\mathbb{Z}/p\mathbb{Z}$.
   Proof: (1) Since G is abelian, $\phi(xy) = (xy)^p = x^p y^p = \phi(x)\phi(y)$.
   (2) Define k_i as n_i/p if $p \mid n_i$ and $k_i = n_i$ otherwise, then $\mathrm{Im}\phi = \bigoplus \mathbb{Z}/k_i\mathbb{Z}$ and $\mathrm{Ker}\phi = (\mathbb{Z}/p\mathbb{Z})^m$ where m is the number of n_i divisible by p.
   (3) Analogously $G/\mathrm{Im}\phi = (\mathbb{Z}/p\mathbb{Z})^m$.

2.3.5

For any group G, define the dual group of G (denoted $\widehat{G}$) to be the set of all homomorphisms from G into the multiplicative group of roots of unity in $\mathbb{C}$ (such homomorphisms are called characters of G). Define a group operation in $\widehat{G}$ by pointwise multiplication of functions: if χ, ψ are homomorphisms from G into the group of roots of unity, then $\chi\psi$ is the homomorphism given by

$$(\chi\psi)(g) = \chi(g)\psi(g)$$

for all $g \in G$, where the latter multiplication takes place in $\mathbb{C}$.

1. Show that this operation on $\widehat{G}$ makes $\widehat{G}$ into an abelian group. In particular, what is the identity element in $\widehat{G}$ and what is the inverse of an element of $\widehat{G}$?
   Remark on notation: It is better to use $\widehat{G}$ only when G is abelian, as $\widehat{G}$ for a nonabelian G often refers to the set of representations of G.
2. Show that if G and H are abelian groups, then

$$\widehat{G \times H} \cong \widehat{G} \times \widehat{H}.$$

3. Compute $\widehat{\mathbb{Z}_n}$ as a group.
4. If G is a finite abelian group, prove that $G \cong \widehat{G}$.
   This result is often phrased as follows: a finite abelian group is self-dual. It implies that the lattice diagram of a finite abelian group is the same when it is turned upside down. Note, however, that there is no natural isomorphism between G and its dual (the isomorphism depends on a choice of a set of generators for G). This is frequently stated in the form: a finite abelian group is non-canonically isomorphic to its dual.
   Proof: (1) The multiplication $\phi\psi : g \mapsto \phi(g)\psi(g)$ is abelian since $\mathbb{C}^\times$ is abelian. The identity is $1 : g \mapsto 1$ and inverse of ψ is $\psi^{-1} : g \mapsto \psi(g)^{-1}$.
   (2) By universal property of products, $\mathrm{Hom}_{\mathbb{Z}}(G \times H, \mathbb{C}^\times) = \mathrm{Hom}(G, \mathbb{C}^\times) \times \mathrm{Hom}(H, \mathbb{C}^\times)$.

(3) A homomorphism $\varphi : \mathbb{Z}/n\mathbb{Z} \to \mathbb{C}^\times$ is uniquely determined by $\varphi(1)$ satisfying $\varphi(1)^n = 1$, which are the nth roots of unity, corresponding to $\mathbb{Z}/n\mathbb{Z}$. Hence $\widehat{\mathbb{Z}_n} = \mathbb{Z}_n$.
(4) Finite abelian groups are the direct sum of $\mathbb{Z}_n$ so by (2) and (3), $\hat{G} \cong G$.

2.3.6

Let I be a nonempty index set and let G_i be a group for each $i \in I$. The restricted direct product, or direct sum, of the groups G_i is the set of elements of the direct product which are the identity in all but finitely many components; that is,

$$\bigoplus_{i\in I} G_i = \left\{ (a_i)_{i\in I} \in \prod_{i\in I} G_i : a_i = 1_i \text{ for all but a finite number of } i \in I \right\}.$$

1. Prove that the restricted product is a normal subgroup of the direct product.
2. Let $I = \mathbb{N}$ and let p_i be the ith integer prime. Show that if $G_i = \mathbb{Z}_{p_i}$, then every element of the restricted direct product of the G_i's has finite order, but $\prod_{i\in I} G_i$ has elements of infinite order. Show that in this example, the restricted product is the torsion subgroup of the direct product.
   Proof: (1) If $(a_i), (b_i) \in G = \bigoplus G_i$ then $a_i b_i^{-1} = 1_i$ for all but finitely many i since $a_i = b_i = 1$, so $(a_i)(b_i)^{-1} \in G$.
   If $(a_i) \in G$ and $(g_i) \in \prod G_i$, then $g_i a_i g_i^{-1} = 1_i$ if $a_i = 1$ which holds for all but finitely many i, so $(g_i)(a_i)(g_i)^{-1} \in G$ and $G \triangleleft \prod G_i$.
   (2) Suppose $(a_i) \in G$ with $a_n = 0 \forall n \geqslant N$, then let $M = \prod_{n<N} p_n$ we have $M(a_i) = 0$ so every element has finite order. However $(1, 1, \cdots) \in \prod G_i$ has infinite order.
   If $(a_i) \notin G$ then suppose $a_{i_k} \neq 1_{i_k}$ for $k = 1, \cdots$. Then for any $n \in \mathbb{Z}$, take $k > n$ then $n \nmid p_{i_k}$ so $a_{i_k}^n \neq 1_{i_k}$. Therefore $(a_i)^n \neq 1$, i.e. (a_i) has infinite order.

2.3.7

Let G and H be two groups and let Z be an (abelian) group equipped with embeddings $i : Z \to G$ and $j : Z \to H$ such that the image $i(Z)$ is contained in the center of G, and the image $j(Z)$ is contained in the center of H.

1. Show that

$$\begin{aligned} \Delta : Z &\longrightarrow G \times H \\ z &\longmapsto \left(i(z), j(z)^{-1}\right) \end{aligned}$$

   defines a natural embedding, and that the image is a normal subgroup of $G \times H$. Denote $G \times_Z H$ to be the quotient

$$G \times_Z H := (G \times H)/\Delta(Z).$$

2. Consider the following example: $G = \mathrm{GL}_2(\mathbb{R})$, $H = \mathbb{C}^\times := \mathbb{C} \setminus \{0\}$ (with multiplication), and $Z := \mathbb{R}^\times = \mathbb{R} \setminus \{0\}$. We hope to relate the product

$$\mathrm{GL}_2(\mathbb{R}) \times_{\mathbb{R}^\times} \mathbb{C}^\times$$

to a unitary group. Consider the Hermitian form $\langle -, - \rangle$ with Hermitian matrix

$$\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$$

defined on $\mathbb{C}^{\oplus 2}$, and the similitude unitary group

$$\mathrm{GU}(2) := \left\{ (g, c) \in \mathrm{GL}_2(\mathbb{C}) \times \mathbb{R}^\times \mid \langle gx, gy \rangle = c\langle x, y \rangle \text{ for all } x, y \in \mathbb{C}^{\oplus 2} \right\}.$$

Show that

$$\mathrm{GL}_2(\mathbb{R}) \times_{\mathbb{R}^\times} \mathbb{C}^\times \cong \mathrm{GU}(2).$$

A version of this isomorphism is used somewhere later in number theory: the construction for the unitary group is easier, yet the construction for GL_2 (or rather its variant) is more subtle. This isomorphism allows one to "transfer" certain structure on $\mathrm{GU}(2)$ to GL_2.

2.3.8

Let H be a normal subgroup of the group G. For each $g \in G$, consider the conjugation on H given by

$$\phi_g : H \longrightarrow H, \qquad h \longmapsto ghg^{-1}.$$

Show that the map $G \to \mathrm{Aut}(H)$ sending $g \mapsto \phi_g$ is a homomorphism. The kernel of this map is

$$C_G(H) := \{ g \in G \mid gh = hg \text{ for all } h \in H \}.$$

This $C_G(H)$ is called the centralizer of H in G.
Proof: $\phi_g \phi_k(h) = \phi_g(khk^{-1}) = gkhk^{-1}g^{-1} = \phi_{gk}(h)$ so $g \mapsto \phi_g$ is a homomorphism.
Note that $\phi_g = \mathrm{id}$ iff $ghg^{-1} = h \forall h \in H$, i.e. $g \in C_G(H)$.

2.3.9

Let H and K be groups and let $\phi : K \to \mathrm{Aut}(H)$ be a homomorphism. Then the following are equivalent:

1. The identity set map between $H \rtimes_\phi K$ and $H \times K$ is a group homomorphism (hence an isomorphism).
2. ϕ is the trivial homomorphism from K into $\mathrm{Aut}(H)$.
3. $K \trianglelefteq H \rtimes_\phi K$.
   Proof: (1)=>(2) If $H \rtimes_\phi K \to H \times K$ is an isomorphism, then for any $(h, k), (h', k') \in H \rtimes_\phi K$, we have $h\phi(k)(h') = hh'$, so $\phi(k) = 1_H$. Therefore $\mathrm{Ker}\phi = K$ and ϕ is trivial.
   (2)=>(1) is analogous.
   (1)=>(3) trivial since $K \triangleleft H \times K$.
   (3)=>(1) For any $k \in K$ and $(h', k') \in H \rtimes_\phi K$, $(h', k')(1, k)(h', k')^{-1}$
   $= (h'\phi(k'k(k')^{-1})(h')^{-1}, k'k(k')^{-1}) \in K$, which implies for any $k \in K$, $\phi(k) = id$.

2.3.10

1. Prove that under any automorphism of D_8, r has at most 2 possible images and s has at most 4

possible images. Deduce that

$$\#\operatorname{Aut}(D_8) \leq 8.$$

2. Use the fact that $D_8 \trianglelefteq D_{16}$ to prove that

$$\operatorname{Aut}(D_8) \cong D_8.$$

What is the center of D_{16}?
Proof: (1) Recall $D_8 = \langle a, b \mid a^2 = b^4 = 1, aba^{-1} = b^{-1} \rangle$. Hence any automorphism $\varphi : D_8 \to D_8$ is determined by $\varphi(a)$ and $\varphi(b)$. Note that b, b^{-1} are the only elements of order 4, so $\varphi(b) = b, b^{-1}$ and both cases are possible, by considering $\varphi = id$ or $\varphi : a \mapsto a, b \mapsto b^{-1}$. Since $\varphi(a) \neq \varphi(b)^2 = b^2$, $\varphi(a)$ can only be a, ab, ab^2, ab^3, and all are possible by $\varphi : a \mapsto ab^k, b \mapsto b$.
Hence there are at most $2 \times 4 = 8$ automorphisms of D_8.
(2) $D_8 \lhd D_{16}$ gives a morphism $\varphi : D_{16} \to \mathrm{Aut}D_8$. Suppose $D_{16} = \langle a, b | a^2 = b^8 = 1, aba = b^{-1} \rangle$ then $D_8 = \langle a, b^2 \rangle$, so $\mathrm{Ker}\varphi = \langle b^4 \rangle$ has order 2. Therefore $|\mathrm{Aut}D_8| = 8$ and $\mathrm{Aut}D_8 = D_{16}/\mathrm{Ker}\varphi \cong D_8$. The center of D_{16} is $\langle b^4 \rangle$.

2.3.11

Assume that K is a cyclic group, H is an arbitrary group, and $\phi_1, \phi_2 : K \to \operatorname{Aut}(H)$ are homomorphisms such that $\phi_1(K)$ and $\phi_2(K)$ are conjugate subgroups of $\operatorname{Aut}(H)$. If K is infinite, then assume that ϕ_1 and ϕ_2 are injective.
Prove, by constructing an explicit isomorphism, that

$$H \rtimes_{\phi_1} K \cong H \rtimes_{\phi_2} K.$$

Challenge question: Why is the condition that ϕ_1 and ϕ_2 be injective when K is infinite needed?
Proof: Suppose $\psi\phi_1(K)\psi^{-1} = \phi_2(K)$, and take $\varphi \in \operatorname{Aut}(K)$ such that $\psi\phi_1(k)\psi^{-1} = \phi_2(\varphi(k))$, then the natural isomorphism will be $(h, k) \mapsto (\psi(h), \varphi(k))$. To find such φ, if $K = \mathbb{Z}$ and ϕ_i are infinite, then $\psi\phi_1(1)\psi^{-1} = \phi_2(1)^{\pm 1}$, so let $\varphi = \pm 1$, and if $K = \mathbb{Z}/n$ then $\psi\phi_1(1)\psi^{-1} = \phi_2(1)^m$ and Chinese remainder theorem gives the desired φ.

2.3.12

Show that for any $n \geq 3$ there are exactly 4 distinct homomorphisms from $\mathbb{Z}_2$ into $\operatorname{Aut}(\mathbb{Z}_{2^n})$. Prove that the resulting semidirect products give 4 nonisomorphic groups of order

$$2^{n+1}.$$

Remark: These four groups, together with the cyclic group and the generalized quaternion group $Q_{2^{n+1}}$, are all the groups of order 2^{n+1} that possess a cyclic subgroup of index 2.
Proof: It suffices to find four elements in $\operatorname{Aut}(\mathbb{Z}/2^n)$ of order one or two, which correspond to elements of $\mathbb{Z}/2^n$ whose square is one. By elementary number theory, there are exactly 4 solutions to $a^2 = 1$ in $\mathbb{Z}/2^n$.
For $\varphi : \mathbb{Z}_2 \to \operatorname{Aut}(\mathbb{Z}/2^n), 1 \mapsto (x \mapsto ax)$ where $a^2 = 1$ in $\mathbb{Z}/2^n$, the semi-direct product gives $G = \langle x, y \mid x^{2^n} = y^2 = 1, yxy = x^a \rangle$. They are not isomorphic, since $a = 1$ gives the only abelian group, and for $a =$

$-1, 2^{n-1}+1, 2^{n-1}-1$, they have respectively $2^n+1, 3, 2^{n-1}+1$ elements of order exactly 2 (since the number is $1+\gcd(1+a, 2^n)$).

2.3.13

Let F be a field, let n be a positive integer, and let G be the group of upper triangular matrices in $\mathrm{GL}_n(F)$.

1. Prove that G is the semidirect product $U \rtimes D$, where U is the set of upper triangular matrices with 1's down the diagonal and D is the set of diagonal matrices in $\mathrm{GL}_n(F)$.
2. Let $n = 2$. Recall that $U \cong F$ and $D \cong F^\times \times F^\times$. Describe the homomorphism from D to $\mathrm{Aut}(U)$ explicitly in terms of these isomorphisms; that is, show how each element of $F^\times \times F^\times$ acts as an automorphism on F.
   Proof: (1) basic linear algebra.
   (2) The homomorphism ϕ satisfy $(UD)\cdot(U'D') = (U\phi(D)(U'))(DD')$ so $DU' = \phi(D)(U')D$ for any D, D', U, U', i.e. $\phi(D): X \mapsto DXD^{-1}$ is the conjugation.
   For $n=2$, $\begin{pmatrix} u & \\ & v \end{pmatrix}\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix}\begin{pmatrix} u^{-1} & \\ & v^{-1} \end{pmatrix} = \begin{pmatrix} 1 & uv^{-1}x \\ & 1 \end{pmatrix}$ hence any element $(u,v) \in F^\times \times F^\times$ acts on F by $x \mapsto uv^{-1}\cdot x$.

2.3.14

Let G be a group acting on sets X and Y. We say that a map $f: X \to Y$ is a G-map or a G-equivariant map if, for every $g \in G$ and $x \in X$,

$$g \cdot f(x) = f(g \cdot x).$$

1. Show that, for $x \in X$, the stabilizer group $\mathrm{Stab}_G(x)$ is a subgroup of $\mathrm{Stab}_G(f(x))$.
2. Consider the situation

$$\phi : X = G/H \longrightarrow Y = G/K$$

for subgroups $H \le K \le G$, sending gH to gK. Show that this map is G-equivariant for the left translation action.

For a point $y = gK \in Y$, show that its preimage $\phi^{-1}(y)$ admits a natural transitive action of gKg^{-1}. Write $\phi^{-1}(y)$ in terms of a coset space of gKg^{-1}.

Proof: (1) If $g \in \mathrm{Stab}_G(x)$ then $gf(x) = f(gx) = f(x)$ so $g \in \mathrm{Stab}_G(f(x))$.
(2) Clearly $\tilde{g}\cdot gK = \tilde{g}\cdot\phi(gH) = (\tilde{g}g)K = \phi(\tilde{g}\cdot gH)$ so it is a G-map.
For $y = gK \in Y$ its preimage is $\{g'H : g'g^{-1} \in K\}$, hence the action is transitive since any $g'H$ is the action of gkg^{-1} on gH where $k = g'g^{-1}$. The fiber corresponds to $\{gkH : k \in K\} \cong (gKg^{-1})/(gHg^{-1})$.

2.3.15

Let H be a subgroup of G and let $N := N_G(H)$ denote its normalizer in G. Show that the coset space G/H carries a natural action of

$$G \times \left(N_G(H)/H\right)$$

given by

$$(g, n)xH = gxn^{-1}H$$

for $g \in G$, $n \in N_G(H)$, and $xH \in G/H$.
Show that this action is transitive. What is the stabilizer subgroup at the identity coset H?
Remarks: Having an action of $G \times (N_G(H)/H)$ is equivalent to saying that we have an action of G and an action of $N_G(H)/H$, and that the two actions commute. When we talk about the stabilizer subgroup, we really mean a group that is explicitly realized as a subgroup of $G \times (N_G(H)/H)$.
Proof: The action is clearly well-defined. For (g, n) and (h, m), we have $(g, n)(h, m)xH = g(hxm^{-1})n^{-1}H$ $= (gh, nm)xH$ so it is a group action.
For any $xH \in G/H$, we have $(x^{-1}, 1)xH = H$ so the action is transitive.
For $xH = H$, $(g, n)H = H \iff gn^{-1} \in H$, so the stabilizer group is just $N_G(H)$, by correspondence g $\in N_G(H) \mapsto (g, gH)$.

2.3.16

Assume that G acts transitively on the finite set A and let H be a normal subgroup of G. Let $O_1, \dots, O_r$ be the distinct orbits of H on A.

1. Prove that G permutes the sets $O_1, \dots, O_r$ in the sense that, for each $g \in G$ and each $i \in \{1, \dots, r\}$, there is a j such that $gO_i = O_j$, where

$$gO = \{g \cdot a \mid a \in O\}.$$

Prove that G is transitive on $\{O_1, \dots, O_r\}$. Deduce that all orbits of H on A have the same cardinality.

2. Prove that if $a \in O_1$, then

$$\#O_1 = [H : H \cap \mathrm{Stab}_G(a)]$$

and prove that

$$r = [G : H\,\mathrm{Stab}_G(a)].$$

The situation considered in this problem will be used quite frequently in studying number theory (in "ramification theory").
Proof: (1) Take $x \in O_i$ and suppose $gx = y \in O_j$. For any $z \in O_j$, suppose $z = hy$ where $h \in H$, then $g^{-1}z$ $= (g^{-1}hg)x \in O_i$ so $z \in gO_i$. Hence $O_j \subset gO_i$ and analogously $O_i \subset g^{-1}O_j$, so $gO_i = O_j$.
Since G is transitive on A, it is also transitive on $\{O_1, \cdots, O_r\}$, and $gO_i = O_j$ implies all orbits have the same cardinality.
(2) Since O_1 is an orbit of H, H acts transitively on O_1, and $\mathrm{Stab}_H(a) = H \cap \mathrm{Stab}_G(a)$. Hence $|O_1| = [H$ $: H \cap \mathrm{Stab}_G(a)]$.
Furthermore, $r = |A|/|O_1| = [G : \mathrm{Stab}_G(a)]/[H : H \cap \mathrm{Stab}_G(a)] = [G : H\mathrm{Stab}_G(a)]$.

2.3.17

Consider the group S_n acting on $\{1, \dots, n\}$. Let $\mathcal{P}$ denote the set of subsets of $\{1, \dots, n\}$. The natural S_n-action on $\{1, \dots, n\}$ induces an action on $\mathcal{P}$ given by: for $\sigma \in S_n$ and $I \subseteq \{1, \dots, n\}$,

$$\sigma(I) := \{\sigma(i) \mid i \in I\}.$$

Find all orbits of $\mathcal{P}$ under this S_n-action. What is the stabilizer of each element of $\mathcal{P}$?
Remark: This generalizes Problem 2.2.9, which may in turn provide some examples for this problem.
Solution: Clearly for any two $I, J \subset [n]$ with $|I| = |J|$, we can find a $\sigma \in S_n$ such that $\sigma(I) = J$, so the orbits are exactly $O_k = \{I \in \mathcal{P} : |I| = k\}$. For $I \in \mathcal{P}$, $\mathrm{Stab}(I) = \{\sigma \in S_n : \sigma(I) = I\}$ which is isomorphic to $S_{|I|} \times S_{n-|I|}$.

2.3.18

Prove that if H is a subgroup of G of index n, then there is a normal subgroup K of G such that $K \leq H$ and

$$[G : K] \leq n!.$$

Proof: Consider the action of G on $G \setminus H$ as $(g, xH) \mapsto gxH$ then it gives a morphism $\varphi : G \to S_n$ with $\mathrm{Ker}\varphi \subset H$ since $gH = H \implies g \in H$. Therefore $K = \mathrm{Ker}\varphi \leqslant H$ and $[G : K] = |\mathrm{Im}\varphi| \leqslant n!$.

2.3.19

Let G be a group with subgroups H and K such that $H \leq K$.

1. Prove that if H is characteristic in K and K is characteristic in G, then H is characteristic in G.
2. Give an example to show that if H is normal in K and K is characteristic in G, then H need not be normal in G.
   Proof: (1) Recall H characteristic means $\phi(H) = H$ for any $\phi \in \mathrm{Aut}(K)$.
   For any $\varphi \in \mathrm{Aut}(G)$, $\varphi(K) = K$ so $\varphi|_K \in \mathrm{Aut}(K)$, and $\varphi(H) = H$. Hence H is characteristic in G.
   (2) Consider $\mathbb{Z}_2 \triangleleft K \triangleleft A_4$, then $K = \{(12)(34), (13)(24), (14)(23), 1\}$ is characteristic in A_4 since automorphism preserves orders, but $\mathbb{Z}_2$ is not normal in A_4.

2.3.20

Let N be a normal subgroup of a group G. Suppose that $\#N = 5$ and that $\#G$ is odd. Prove that N is contained in the center of G.
Proof: Consider $\varphi : G \to \mathrm{Aut}(N), g \mapsto (x \mapsto gxg^{-1})$ then $N < C(G)$ iff $\mathrm{Ker}\varphi = G$. Since $|\mathrm{Im}\varphi| = |G|/|\mathrm{Ker}\varphi|$ is odd and a factor of $|\mathrm{Aut}N|$, where $\mathrm{Aut}N = \mathrm{Aut}\mathbb{Z}_5 = \mathbb{Z}_4$, we know that $\mathrm{Im}\varphi = 0$ and $\mathrm{Ker}\varphi = G$.

2.3.21

1. Suppose that a group G operates transitively on a set S, and that H is the stabilizer of an element $s_0 \in S$. Consider the action of G on $S \times S$ defined by

$$g(s_1, s_2) = (gs_1, gs_2).$$

Establish a bijective correspondence between double cosets of H in G and G-orbits in $S \times S$.

2. Work out the correspondence explicitly for the case that G is the dihedral group D_5 and S is the set of vertices of a 5-gon.
   Proof: (1) For any $HgH \in H \setminus G/H$, define $A_g = G \cdot (s_0, gs_0)$, then each A_g is a G-orbit. If (s_1, s_2) $\in S \times S$, then suppose $s_i = g_i s_0$ and $g_1^{-1} g_2 \in HgH$, then $g_1^{-1} g_2 = h_1 g h_2$ implies (s_1, s_2) $= g_1(s_0, h_1 g h_2 s_0) = g_1 h_1^{-1} \cdot (s_0, gs_0)$. Therefore A_g covers all orbits. If $(s_0, gs_0) = x \cdot (s_0, \tilde{g} s_0)$, then $xs_0 = s_0 \implies x \in H$ and $gs_0 = x\tilde{g}s_0$ implies $g = x\tilde{g}y$ for some $x, y \in H$, so $HgH = H\tilde{g}H$.
   (2) The double cosets are

$$\begin{aligned} H &= \{1, t\}, \\ HrH &= \{r, r^4, rt, r^4 t\}, \\ Hr^2H &= \{r^2, r^3, r^2 t, r^3 t\}. \end{aligned}$$

The G-orbits are correspondingly, $\{(i, i)\}$, $\{(i, i \pm 1)\}$, $\{(i, i \pm 2)\}$.

2.3.22

Find a composition series for A_4. Deduce that A_4 is solvable.
Proof: Let $K = \langle (12)(34), (13)(24), (14)(23), e \rangle$ then $K \triangleleft A_4$ and A_4/K has index 3 hence simple and abelian. Clearly $K = \mathbb{Z}_2 \times \mathbb{Z}_2$ so A_4 has solvable series $1 \triangleleft \mathbb{Z}_2 \triangleleft K \triangleleft A_4$.

2.3.23

Prove that A_n contains a subgroup isomorphic to S_{n-2} for each $n \geq 3$.
Proof: S_{n-2} acts on $\{1, \cdots, n\}$ in the following way: for each σ, it maps $1, \cdots, n-2$ by $k \mapsto \sigma(k)$, and then swaps $n-1, n$ if σ is odd. Then the action is an even permutation in S_n, hence an embedding $S_{n-2} \hookrightarrow A_n$.

2.3.24

This problem combines the left translation, the right translation, and the conjugation action of G on itself.
Fix a group G for this discussion.

1. Show that there is an action of $G \times G$ on G given as follows: for $(g, h) \in G \times G$, define a bijection

$$\begin{aligned} \Phi_{g,h} &: G \longrightarrow G \\ \Phi_{g,h}(x) &= gxh^{-1}. \end{aligned}$$

2. Show that the stabilizer group of this $G \times G$-action at each element $x \in G$ is isomorphic to G. Note: these subgroups are isomorphic but are not the same as subgroups.
3. Show that the left translation, right translation, and adjoint actions may be viewed as restrictions of this $G \times G$-action to certain subgroups of $G \times G$.
   Proof: (1) It is clearly a group action since $(g, h)(k, l)x = gkxl^{-1}h^{-1} = (gk, hl)x$.
   (2) Note that $\mathrm{Stab}_{G \times G}(x) = \{(g, gx) : g \in G\}$ which is isomorphic to G.

(3) Left translation is restriction to $G \times e$, right translation is to $e \times G$ and adjoint action to the diagonal $\Delta \subset G \times G$.

2.4. More Difficult Questions

2.4.1

Let p be a prime. Write the following abelian groups additively.

1. Consider the group

$$G = \mathbb{Z}_{p^{m_1}} \times \cdots \times \mathbb{Z}_{p^{m_r}}$$

with $m_1 \leq \cdots \leq m_r$. Compute the order of the p^N-torsion subgroup of G:

$$G[p^N] := \{x \in G \mid p^N x = 0\}.$$

2. Let

$$H = \mathbb{Z}_{p^{n_1}} \times \cdots \times \mathbb{Z}_{p^{n_s}}$$

be another abelian group with $n_1 \leq \cdots \leq n_s$. Show that

$$G \cong H$$

if and only if $r = s$ and $m_i = n_i$ for each $i = 1, \dots, r$.

Proof: (1) $p^N x = 0$ iff $p^N x_i = 0$ for each x_i in $x = (x_1, \cdots, x_r)$. Hence $G[p^N] = \bigoplus \mathbb{Z}_{p^{m_i}}[p^N]$ where $\mathbb{Z}_{p^M}[p^N] = \mathbb{Z}/p^{\min\{M,N\}}\mathbb{Z}$, so $G[p^N] = \bigoplus \mathbb{Z}/p^{\min\{m_i,N\}}\mathbb{Z}$.

(2) If $G \cong H$, then the index of $G[p^n]$ gives the number of m_i such that $m_i \geqslant n$. Hence let $n = 1$ we have $r = s$, and by increasing n we obtain $m_i = n_i$.

2.4.2

Let S_n be the group of permutations of $\{1, \dots, n\}$. Let $\sigma \in S_n$ be the permutation

$$\sigma = (1, n)(2, n-1) \cdots (k, n-k+1) \cdots \left(\left\lfloor \frac{n-1}{2} \right\rfloor, \left\lceil \frac{n+2}{2} \right\rceil \right).$$

Prove that the centralizer $Z_{S_n}(\sigma)$ is isomorphic to

$$S_{\lfloor n/2 \rfloor} \ltimes (\mathbb{Z}_2)^{\lfloor n/2 \rfloor}.$$

Proof: Note that σ consists of $\lfloor n/2 \rfloor$ disjoint 2-cycles. Hence for any permutation $\tau \in S_n$ that first permutes the $\lfloor n/2 \rfloor$ 2-cycles, then choose to interchange the two elements of some of the 2-cycles, τ commutes with σ. Conversely, if $\tau \in Z_{S_n}(\sigma)$, then for any $i < \frac{n-1}{2}$, $\sigma(\tau(i)) = \tau(\sigma(i)) = \tau(n+1-i)$ so τ takes $\{i, n+1-i\}$ to some pair $\{j, n+1-j\}$. Hence τ first permutes the $\lfloor n/2 \rfloor$ 2-cycles $\{j, n+1-j\}$, then permutes in each 2-cycle. Therefore these are exactly all elements of $Z_{S_n}(\sigma)$, and they consist of $S_{\lfloor n/2 \rfloor}$ acting on the

permutations of $\lfloor n/2 \rfloor$ 2-cycles, i.e. $S_{\lfloor n/2 \rfloor} \ltimes (\mathbb{Z}_2)^{\lfloor n/2 \rfloor}$.

2.4.3

Let G be a group of order n with n odd. Prove that the left translation action gives a homomorphism

$$G \longrightarrow A_n.$$

Proof: Left translation gives $\varphi : G \to S_n$ so $\psi = \mathrm{sgn} \circ \varphi : G \to \{\pm 1\}$. If $\mathrm{Ker}\psi \neq G$ then $\mathrm{Im}\psi = G/\mathrm{Ker}\psi \neq e$ implies ψ is surjective, so $|G| = 2|\mathrm{Ker}$
$psi|$ a contradiction. Hence $\mathrm{Ker}\psi = G$ and $\mathrm{Im}\varphi \subset A_n$. Hence φ gives $G \to A_n$.

2.4.4

Let G be a finitely generated abelian group. The classification theorem says that G is isomorphic to a product of "standard" abelian groups, but such an isomorphism is not "canonical." We discuss this matter here. We write the group operation in G additively. Let us assume that

$$G \cong \mathbb{Z} \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r}$$

for positive integers $n_1 \geq 2$ and $n_i \mid n_{i+1}$ for $i = 1, \ldots, r-1$.

1. Define

$$G_{\mathrm{tor}} := \{g \in G \mid m \cdot g = 0 \text{ for some } m \in \mathbb{N}\}.$$

This is the torsion subgroup of G. Show that if G is isomorphic to $\mathbb{Z} \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r}$, then

$$G/G_{\mathrm{tor}} \cong \mathbb{Z}.$$

This G_{tor} is a canonical subgroup, and $G/G_{\mathrm{tor}} \cong \mathbb{Z}$ is a canonical torsion-free quotient.

2. Describe all injective homomorphisms $\phi : \mathbb{Z} \to G$ such that $G/\phi(\mathbb{Z})$ is isomorphic to G_{tor}. How many are there?
   Show that every such ϕ induces an isomorphism

$$\widetilde{\phi} : \mathbb{Z} \times G_{\mathrm{tor}} \longrightarrow G.$$

(But, of course, there is no canonical such choice.)
Proof: (1) An isomorphism keeps torsion elements, and the torsion elements of $\mathbb{Z} \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r}$ is $0 \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r}$, so $G/G_{\mathrm{tor}} \cong \mathbb{Z}$.
(2) For any torsion-free element $g \in G$ identified in $\{\pm 1\} \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r}$, consider $\phi_g : 1 \mapsto g$, then $\phi(\mathbb{Z})$ is infinite cyclic, and for any $h \in G$ identified with $(m, x_1, \cdots, x_r)$, $h \pm mg \in G_{\mathrm{tor}}$, so $G/\phi(\mathbb{Z}) \cong G_{\mathrm{tor}}$. Hence there are in total $2m_1 \cdots m_r$ such homomorphisms ϕ, and every such ϕ induces $\mathbb{Z} \times G_{\mathrm{tor}} \to G$ by (m, h) $\mapsto mg + h$ which is an isomorphism.

2.4.5

(Yau contest 2017)
Let A be a finite abelian group and let $\phi : A \to A$ be an endomorphism. Put

$$A_{\text{nil}} := \{x \in A \mid \phi^k(x) = 0 \text{ for some } k \geq 1\}.$$

Show that there is a unique subgroup A_0 of A such that ϕ restricts to an automorphism of A_0 and

$$A = A_0 \times A_{\text{nil}}.$$

This problem seems a little strange if one sees it for the first time. A good example is the following: let p be an odd prime and $r \in \mathbb{N}$, and let

$$A = \mathbb{Z}_{p^r}^4.$$

The homomorphism ϕ is given by the matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & p \end{pmatrix}.$$

The decomposition $A = A_0 \times A_{\text{nil}}$ is used, for example, in p-adic number theory.
Proof: A is a Noetherian and Artinian $\mathbb{Z}$-module, so $\mathrm{Ker}\phi^k \subset \mathrm{Ker}\phi^{k+1}$ implies for some k, $A_{\text{nil}} = \bigcup \mathrm{Ker}\phi^i$ $= \mathrm{Ker}\phi^k = \mathrm{Ker}\phi^{k+1} = \cdots$ and likewise $\mathrm{Im}\phi^k$ will stabilize. Take k large enough, and let $A_0 = \mathrm{Im}\phi^k$, then A_0 $\cap A_{\text{nil}} = \emptyset$ and $A/A_0 \cong A_{\text{nil}}$, so we have a canonical isomorphism $A_0 \oplus A_{\text{nil}} \to A$. A_0 is also unique since $\mathrm{Im}\phi^k = \phi^k(A_0) + \phi^k(A_{\text{nil}}) = A_0$.

2.4.6

Let G be a finite $\mathbb{Z}$-module (that is, a finite abelian group with additive group law) with a bilinear, (strongly) alternating, and nondegenerate pairing

$$\ell : G \times G \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

Here "(strongly) alternating" means that for every $a \in G$,

$$\ell(a, a) = 0;$$

"nondegenerate" means that for every nonzero $a \in G$, there is a $b \in G$ such that $\ell(a, b) \neq 0$. Show, in steps, the following statement:

$$(\mathrm{S})\text{: } G \text{ is isomorphic to } H_1 \oplus H_2$$

for some finite abelian groups $H_1 \cong H_2$ such that

$$\ell\big|_{H_i \times H_i} = 0$$

for each $i = 1, 2$.

1. For every $a \in G$, write $o(a)$ for the order of a and let $\ell_a : G \to \mathbb{Q}/\mathbb{Z}$ be the homomorphism

$$\ell_a(b) = \ell(a, b).$$

Show that the image of ℓ_a is

$$o(a)^{-1}\mathbb{Z}/\mathbb{Z}.$$

2. Show that G has a pair of elements a, b with the following properties:
   - $o(a)$ is maximal in the sense that, for any $x \in G$,

$$o(x) \mid o(a).$$

   -

$$\ell(a, b) = o(a)^{-1} \pmod{\mathbb{Z}}.$$

   - $o(a) = o(b)$.

   We call the subgroup

$$\langle a, b\rangle := \mathbb{Z}a + \mathbb{Z}b$$

a maximal hyperbolic subgroup of G.

3. Let $\langle a, b\rangle$ be a maximal hyperbolic subgroup of G. Let G' be the orthogonal complement of $\langle a, b\rangle$, consisting of elements $x \in G$ such that

$$\ell(x, c) = 0$$

for all $c \in \langle a, b\rangle$. Show that G is a direct sum as follows:

$$G = \mathbb{Z}a + \mathbb{Z}b + G'.$$

4. Finish the proof of (S) by induction.

   One origin of such a group G is the so-called Tate-Shafarevich group of an elliptic curve over a number field. Such a group comes equipped with a perfect alternating pairing if it is known to be finite.

   Proof: (1) If $q = \ell(a, b) \in \mathrm{Im}\ell_a$, then $o(a)q = \ell(o(a)a, b) = 0$ so $q \in o(a)^{-1}\mathbb{Z}/\mathbb{Z}$. If $\mathrm{Im}\ell_a \subsetneq o(a)^{-1}\mathbb{Z}/\mathbb{Z}$ suppose $\mathrm{Im}\ell_a = m^{-1}\mathbb{Z}/\mathbb{Z}$ where $m \mid o(a)$, then $\ell(ma, b) = m\ell(a, b) = 0$ for any b, so $ma = 0$ and $o(a) \leqslant m$ a contradiction. Therefore $\mathrm{Im}\ell_a = o(a)^{-1}\mathbb{Z}/\mathbb{Z}$.

   (2) Take any a such that $o(a)$ is maximal (this is always possible for a finite group), and by (1) take b such that $\ell(a, b) = o(a)^{-1}$. Then $\frac{o(b)}{o(a)} = \ell(a, o(b)b) = 0$ implies $o(b) = o(a)$.

   (3) If $g' \in G' \cap (\mathbb{Z}a + \mathbb{Z}b)$, suppose $g' = ua + vb$ then $\frac{u}{o(a)} = \ell(g', b) = 0$ and $\frac{v}{o(a)} = \ell(g', a) = 0$ so $g' = 0$.

If $g \in G$ let $u = o(a)\ell(g', b)$ and $v = o(a)\ell(g', a)$, then $g' = g - ua - vb$ satisfy $\ell(g', a) = \ell(g', b) = 0$ so $g' \in G'$. Therefore $G = (\mathbb{Z}a + \mathbb{Z}b) \oplus G'$.
(4) By induction we can write G as the direct sum $G = \bigoplus(\mathbb{Z}a_i + \mathbb{Z}b_i)$. Let $H_1 = \bigoplus \mathbb{Z}a_i$ and $H_2 = \bigoplus \mathbb{Z}b_i$, then $o(a_i) = o(b_i)$ implies $H_1 \cong H_2$, and by definition $\ell|_{H_i \times H_i} = 0$.

2.4.7

This exercise shows that for $n \neq 6$, every automorphism of S_n is inner. Fix an integer $n \geq 2$ with $n \neq 6$.

1. Prove that the automorphism group of a group G permutes the conjugacy classes of G; that is, for each $\sigma \in \mathrm{Aut}(G)$ and each conjugacy class C of G, the set $\sigma(C)$ is also a conjugacy class of G.
2. Let C be the conjugacy class of transpositions in S_n, and let C' be the conjugacy class of any element of order 2 in S_n that is not a transposition. Prove that

$$|C| \neq |C'|.$$

Here we use $n \neq 6$. Deduce that any automorphism of S_n sends transpositions to transpositions.

3. Prove that, for each $\sigma \in \mathrm{Aut}(S_n)$,

$$\sigma : (12) \longmapsto (ab_2), \qquad \sigma : (13) \longmapsto (ab_3), \qquad \dots, \sigma : (1n) \longmapsto (ab_n)$$

for some distinct integers $a, b_2, b_3, \dots, b_n \in \{1, \dots, n\}$.

4. Since $(12), (13), \dots, (1n)$ generate S_n, deduce that any automorphism of S_n is uniquely determined by its action on these elements. Use (3) to show that S_n has at most $n!$ automorphisms and conclude that

$$\mathrm{Aut}(S_n) = \mathrm{Inn}(S_n)$$

for $n \neq 6$.
Proof: (1) If C is a conjugacy class of G, i.e. $gCg^{-1} = C \forall G \in G$, then for any automorphism $\phi : G \to G$ and $g \in G$, take $\phi(h) = g$ then $g\phi(c)g^{-1} = \phi(hch^{-1}) \in \phi(C)$ for any $\phi(c) \in \phi(C)$, and likewise $\phi(C)$ is transitive under conjugacy action, hence automorphisms permutes conjugacy classes.
(2) Suppose C' consists of elements with k 2-cycles, then $|C'| = \frac{1}{k!}\binom{n}{2}\binom{n-2}{2}\cdots\binom{n-2k+2}{2} = \frac{n!}{k!(n-2k)!2^k}$. When $n \neq 6$, $\binom{n}{2} \neq \frac{n!}{2^k k!(n-2k)!}$ so $|C| \neq |C'|$.
(3) Since σ preserves order and permutes conjugacy classes, by (2) transpositions are mapped to transpositions under σ. Suppose $\sigma : (12) \mapsto (ab_2)$ then $\sigma((12k))$ has order 3, so $\sigma((1k))$ and $\sigma((12))$ have a common element for each k. Therefore $\sigma : (12) \mapsto (ab_2), (13) \mapsto (ab_3), \cdots, (1n) \mapsto (ab_n)$.
(4) By (3) $a, b_2, \cdots, b_n$, i.e. a permutation of $\{1, \cdots, n\}$ completely determine $\sigma \in \mathrm{Aut}S_n$. Hence there are at most $n!$ automorphisms, and we already have $n!$ distinct automorphisms in $\mathrm{Inn}S_n$ since the center of S_n is trivial. Therefore $\mathrm{Aut}S_n = \mathrm{Inn}S_n$ for $n \neq 6$.

2.4.8

Prove that if a group G is finitely generated, then any subgroup of finite index in G is finitely generated.
Proof: Suppose $G = \langle a_1, \cdots, a_n \rangle$, and $H < G$ with $[G : H] = \{g_1, \cdots, g_m\}$ where $g_1 = 1$. Suppose $sg_j = g_i h_{s,j}$ where $h_{s,j} \in H$, for each $s \in \{a_1^{\pm 1}, \cdots, a_n^{\pm 1}\}$, then for any $h = s_1 \cdots s_r \in H$, we have $s_r = s_r g_1 = g_i h_{s_r,1}$, $s_{r-1} g_i = g_j h_{s_{r-1},i}$ etc. Repeatedly apply the above process, we write $h = g_i \prod h_{s,j}$. Therefore $g_i = 1$ and every element $h \in H$ is generated by the finite set $\{h_{s,j}\}$.

2.4.9

Let K and L be groups, let n be a positive integer, let $\rho : K \to S_n$ be a homomorphism, and let H be the direct product of n copies of L. Then there is a natural homomorphism

$$\psi : S_n \longrightarrow \mathrm{Aut}(H)$$

by permuting the n factors of H. The composition $\psi \circ \rho$ is a homomorphism from K into $\mathrm{Aut}(H)$. The wreath product of L by K is the semidirect product $H \rtimes K$ with respect to this homomorphism and is denoted by $L \wr K$. This wreath product depends on the choice of permutation representation ρ of K and, of course, on the number n; if none is given explicitly, ρ is assumed to be the left regular representation of K.

1. Assume K and L are finite groups and ρ is the left regular representation of K. Find $\#(L \wr K)$ in terms of $\#K$ and $\#L$.
2. Let p be a prime, let $K = L = \mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$, and let ρ be the left regular representation of K. Prove that $\mathbb{Z}_p \wr \mathbb{Z}_p$ is a nonabelian group of order

$$p^{p+1}$$

and is isomorphic to a Sylow p-subgroup of S_{p^2} (the permutation group of p^2 elements).

3. Show that $\mathbb{Z}_2 \wr S_n$ (the hyperoctahedral group) is the symmetry group of the n-dimensional cube. The action of S_n on $\{1, \ldots, n\}$ is the usual one.
   Proof:

2.4.10

Let G be a finite group of composite order n with the property that G has a subgroup of order k for each positive integer k dividing n. Prove that G is not simple.
Proof:

2.4.11

1. Recall that a proper subgroup M of G is called maximal if, whenever $M \leq H \leq G$, either $H = M$ or $H = G$. Prove that if M is a maximal subgroup of G, then either $N_G(M) = M$ or $N_G(M) = G$.
   Deduce that if M is a maximal subgroup of G that is not normal in G, then the number of nonidentity elements of G that are contained in conjugates of M is at most $(\#M - 1)[G : M]$.
2. Assume H is a proper subgroup of the finite group G. Prove that

$$G \neq \bigcup_{g \in G} gHg^{-1};$$

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that is, $G$ is not the union of the conjugates of any proper subgroup.
**Remark:** This problem has the following application later in number theory: Let $L$ be a finite extension of a number field $K$. Then there exist infinitely many unramified places $v$ of $K$ such that every place of $L$ over $v$ has degree greater than 1 over $v$.
```

Proof:

2.4.12

(Cohen-Lenstra density question)

1. Let p be a prime. Compute the order of the automorphism group of

$$\mathbb{Z}_{p^{n_1}} \times \cdots \times \mathbb{Z}_{p^{n_r}}$$

with $n_1 \leq \cdots \leq n_r$.

2. Define

$$(p)_r := \prod_{i=1}^{r} (1 - p^{-i}).$$

Show that

$$\sum_{\substack{G \text{ is } p\text{-abelian} \\ \#G \leq p^r}} \frac{1}{\#\operatorname{Aut}(G)} = \frac{1}{(p)_r},$$

where the sum takes over all finite abelian groups that have order $\#G \leq p^r$. (I have not tried this problem myself, but I have checked it for $r = 2, 3$ by hand; I do not know if there is a nice proof.)

Taking the limit shows that

$$\sum_{G \text{ is } p\text{-abelian}} \frac{1}{\#\operatorname{Aut}(G)} = \frac{1}{(p)_\infty}.$$

Remark: The background of this question is the so-called Cohen-Lenstra heuristic. Consider all imaginary quadratic fields

$$F = \mathbb{Q}(\sqrt{-d})$$

with d a square-free positive integer. Its ring of integers is

$$\mathcal{O}_F = \begin{cases} \mathbb{Z}[\sqrt{-d}], & -d \equiv 2,3 \pmod 4, \\ \mathbb{Z}\left[\dfrac{1+\sqrt{-d}}{2}\right], & -d \equiv 1 \pmod 4. \end{cases}$$

Then there is a question of whether $\mathcal{O}_F$ has a property that every element admits a unique factorization into primes, just like in $\mathbb{Z}$. This is of course not correct in general. To characterize the failure of this, one may naturally introduce a finite abelian group called the ideal class group $\mathrm{Cl}(\mathcal{O}_F)$. The group $\mathrm{Cl}(\mathcal{O}_F)$ is trivial if and only if $\mathcal{O}_F$ admits the unique factorization property. For imaginary quadratic fields F, it is known (Gauss' conjecture) that there are only 9 imaginary quadratic fields.

Cohen-Lenstra says that for any finite abelian group G of p-power order,

$$\lim_{D\to\infty} \frac{\#\left\{1 < d \le D \text{ square-free} \mid \mathrm{Cl}\left(\mathcal{O}_{\mathbb{Q}(\sqrt{-d})}\right)[p^\infty] \cong G\right\}}{\#\left\{1 < d \le D \text{ square-free}\right\}}$$

is proportional to

$$\frac{1}{\#\operatorname{Aut}(G)}.$$

Here, $A[p^\infty]$ means the p-power torsion subgroup of the corresponding abelian group, or the p-Sylow subgroup. This is the “correct” randomness: namely, the ideal class group is a “random” finite abelian group, weighted by the size of its automorphism group.

For real quadratic fields, there is also a similar conjecture, but with a more complicated heuristic; namely, the ideal class group is supposed to be a random group quotient by a random cyclic subgroup. In particular, conjecturally, around 75.446% of real quadratic fields have the unique factorization property. It is not even known whether there are infinitely many such real quadratic fields (known as Gauss' conjecture on real quadratic fields).

Proof: (1) Automorphisms preserve order, so $1 \in \mathbb{Z}_{p^s}$ can only be mapped to some other group $\mathbb{Z}_{p^t}$ with $t \geqslant s$. Hence by induction, we only need to consider $\operatorname{Aut}((\mathbb{Z}_{p^r})^n)$, then $\operatorname{Aut}\left(\bigoplus \mathbb{Z}_{p^{n_i}}\right) = \bigoplus \operatorname{Aut}((\mathbb{Z}_{p^r})^{\#\{i:n_i=r\}})$. Recall $\operatorname{Aut}(\mathbb{Z}_{p^a})^n = \mathrm{GL}_n(\mathbb{Z}/p^a\mathbb{Z})$, which has order $|\mathrm{Aut}G| = p^{\sum_{i,l=1}^{r} \min(n_i,n_l)} \prod_{j\geq 1}(p)_{k_j}$ where $k_j = \#\{i : n_i = j\}$.

(2)

2.4.13

(Alibaba 2022)

Let $G_1, \dots, G_n$ be nonabelian simple groups for some integer $n \geq 2$, and let H be a subgroup of $G_1 \times \cdots \times G_n$ such that the projection homomorphism

$$H \longrightarrow G_i \times G_j$$

is surjective for every pair of indices $i < j$. Show that

$$H = G_1 \times \cdots \times G_n.$$

Proof: